

WaveLens Lite — Progress Report

Convo AI Hackathon 2026 | Team: PiX.lab

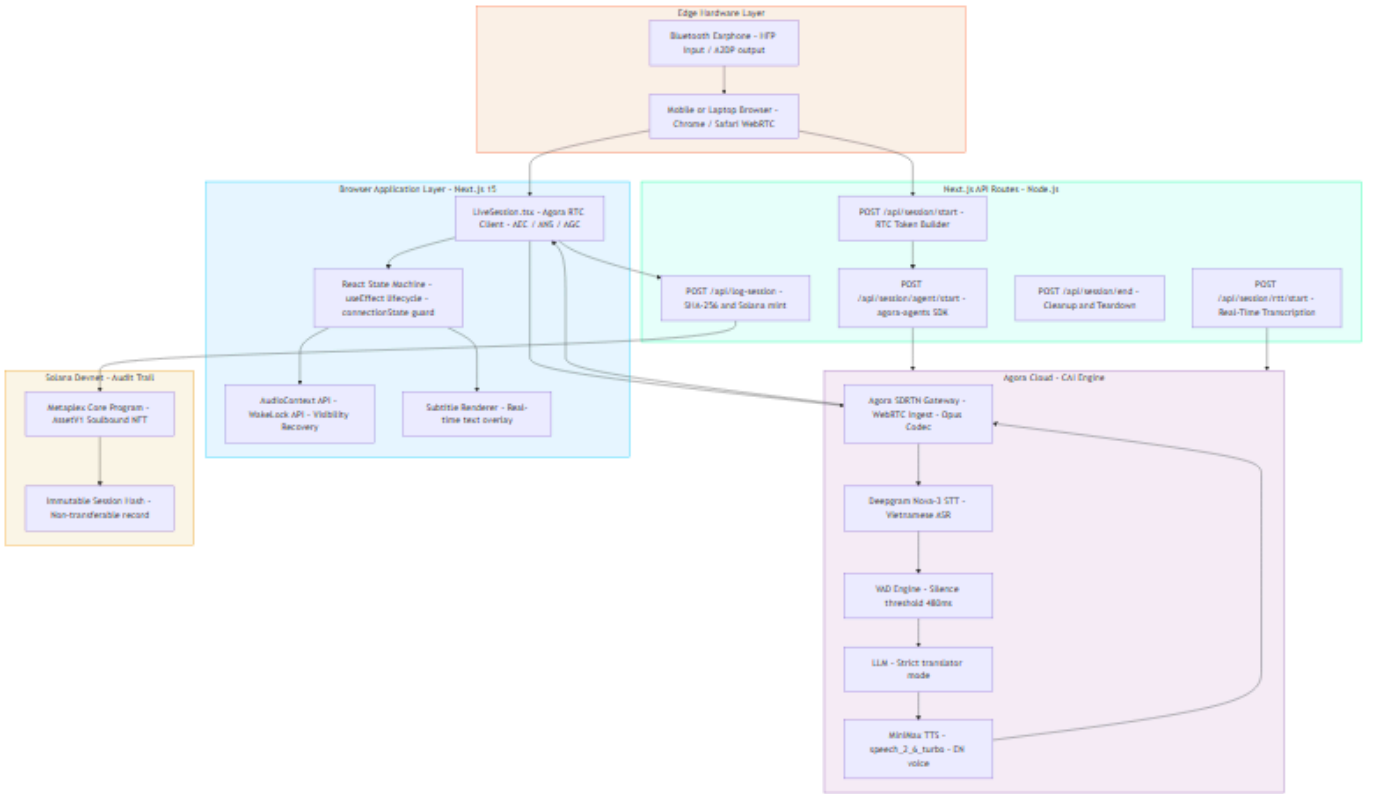
Deployment Context: All screenshots are captured from a real, fully operational instance running on local hardware — a laptop paired with a mobile device over Bluetooth earphone. The system is **not yet deployed to a cloud server** — this phase validates the complete end-to-end pipeline on physical hardware before production deployment.

Executive Summary

WaveLens Lite is a production-grade, real-time multilingual voice translation system purpose-built for industrial edge environments — specifically Vietnamese maritime port operations. It captures speech through a connected Bluetooth earphone, routes it through the **Agora Conversational AI Engine (CAI)**, and returns synthesized translated audio and live text subtitles in under 2 seconds. Every session is cryptographically signed and anchored to the Solana blockchain as an immutable compliance audit record.

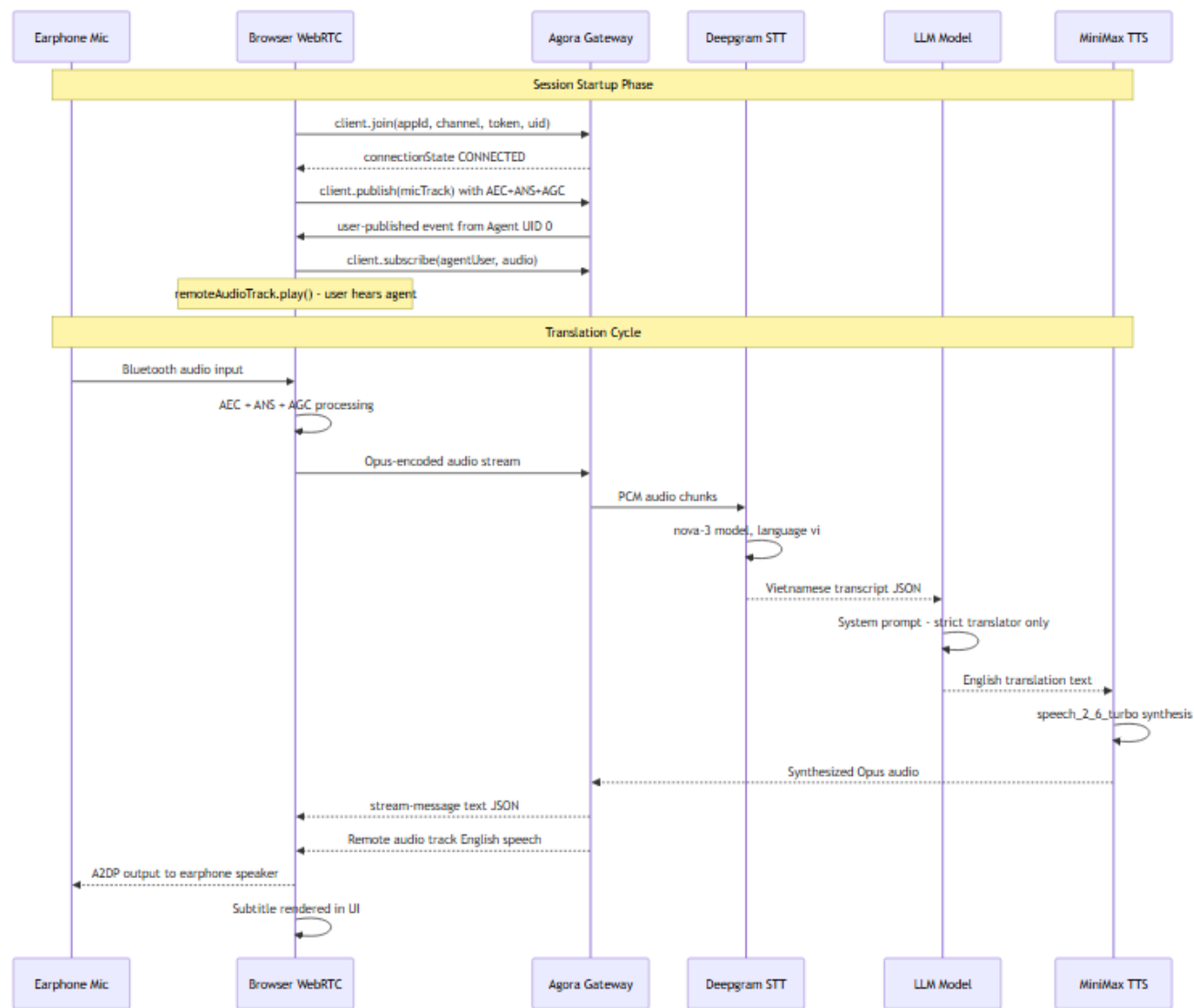
This document provides an expert, end-to-end architectural analysis from the perspective of a production AI Systems Engineer, detailing every subsystem, its engineering rationale, known failure modes, and implemented mitigations.

1. System-Level Overview



2. Dual-Channel Real-Time Translation Pipeline

The core of WaveLens is a bidirectional audio/text pipeline over the **Agora SDRTN**. Two parallel channels run simultaneously: the audio channel carries synthesized translated speech, and the data channel carries UTF-8 JSON text chunks for real-time subtitle rendering.

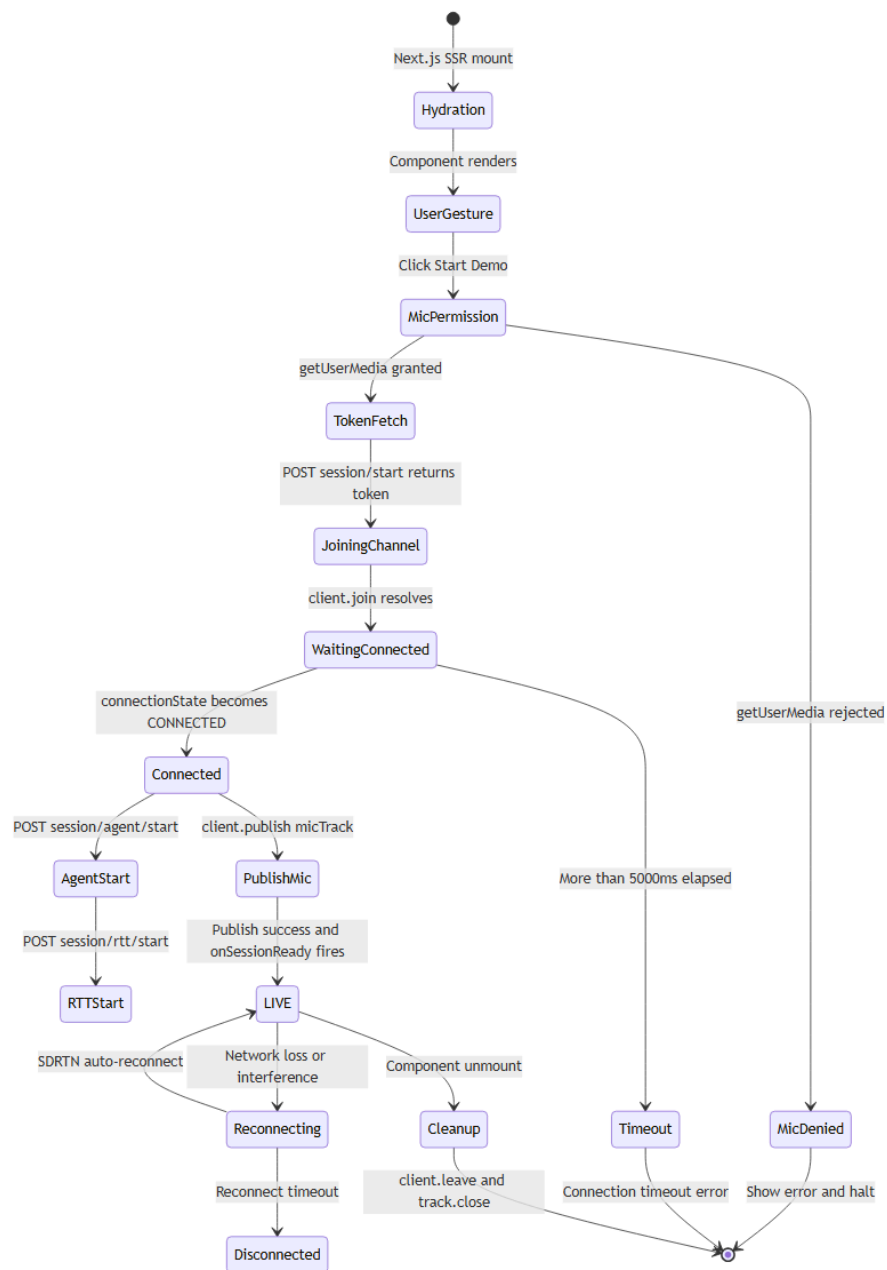


Engineering Rationale

Constraint	Why It Matters
<code>AEC: true</code>	Prevents the LLM from hearing its own translated output and re-translating it, causing a feedback loop
<code>ANS: true</code>	Da Nang port machinery produces 85-110 dB noise; suppression maintains clean VAD thresholds
<code>AGC: true</code>	Normalizes volume across workers shouting in wind vs. speaking quietly in a confined compartment
Agora SDRTN	Purpose-built real-time relay network with lower jitter than commodity WebRTC in high-packet-loss environments
Dual data channel	Text stream runs parallel to audio — if audio is too noisy, subtitles provide a visual fallback

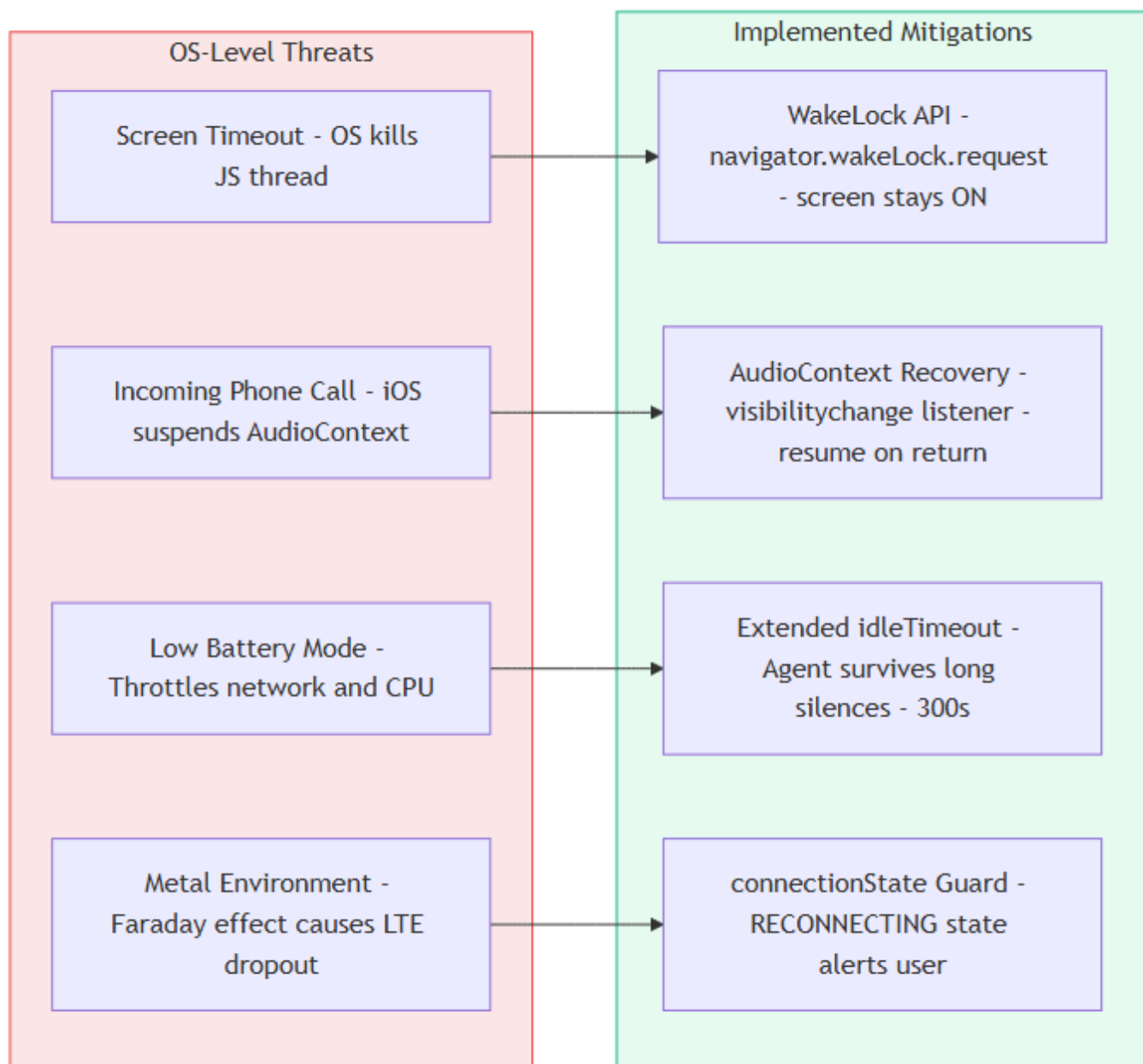
3. State Machine Lifecycle

This is the most critical engineering layer. A single broken state transition is the root cause of the `INVALID_OPERATION` publish error — the most common failure mode in real-world Agora deployments.

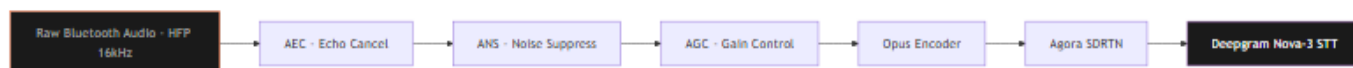


4. Mobile Hardware Resilience Architecture

Industrial deployments face OS-level threats invisible in a development environment. Four threat categories are mitigated at the browser layer.

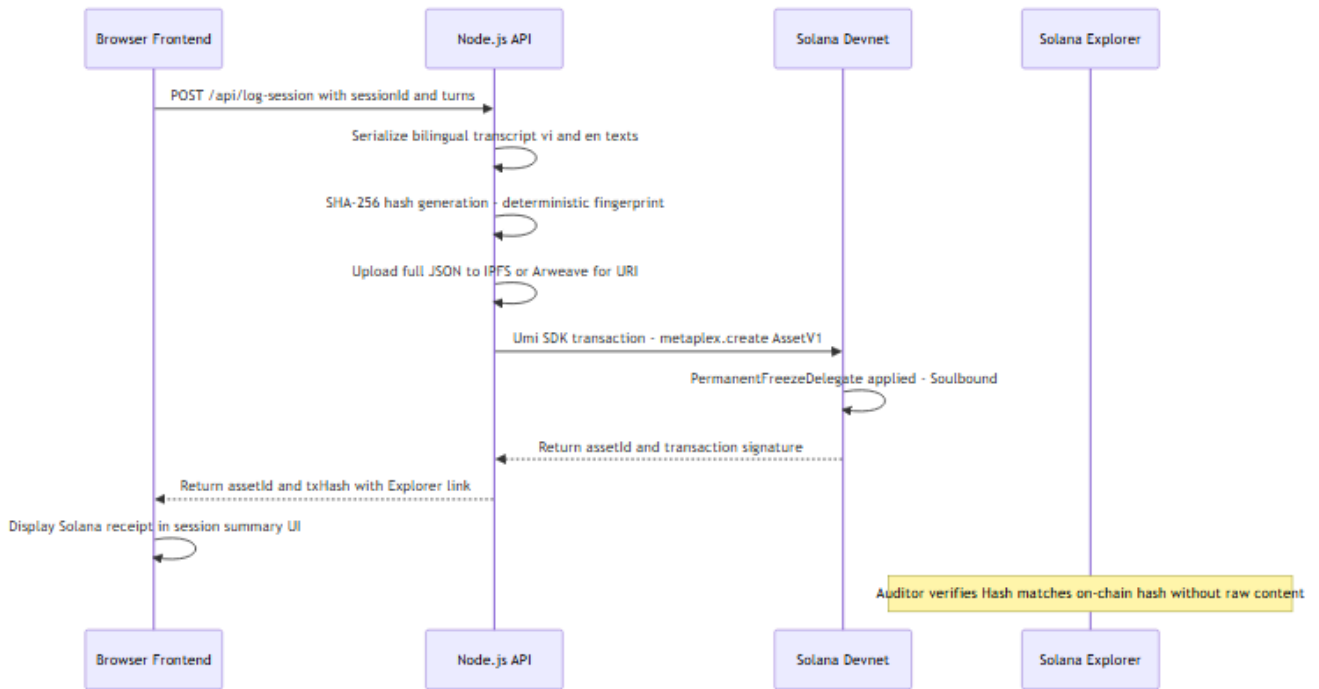


Microphone Signal Processing Stack



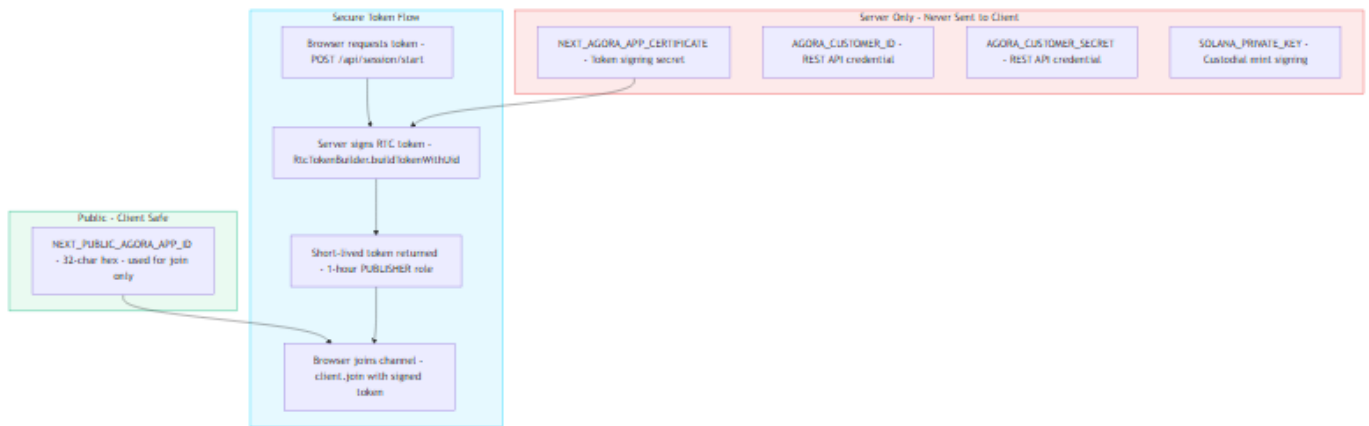
5. Blockchain Audit Trail — Solana / Metaplex Core

Industrial safety compliance mandates immutable, tamper-proof logging. Storing raw transcripts on-chain is cost-prohibitive. WaveLens uses a **cryptographic hash commitment** model.



Note: Privacy-preserving design: Only the SHA-256 hash is stored on-chain. The full bilingual transcript remains off-chain (IPFS/Arweave, access-controlled). This satisfies both audit requirements and data sovereignty concerns common in Vietnamese maritime operations.

6. Security Architecture



7. Technology Stack

Layer	Technology	Rationale
Frontend Framework	Next.js 15 + Turbopack	App Router + Server Components; fast HMR for development
UI Language	TypeScript + React 18	Strict type safety for complex async state machines
RTC SDK	agora-rtc-sdk-ng	Browser-native WebRTC with SDRTN global relay
AI Agent SDK	agora-agents	First-party SDK for full CAI pipeline orchestration

STT	Deepgram Nova-3	Best available Vietnamese ASR; real-time streaming
LLM	Large Language Model	Best cost/latency ratio for translation at under 1 second
TTS	MiniMax speech_2_6_turbo	Sub-200ms synthesis; natural English voice output
Token Auth	agora-token RtcTokenBuilder	Server-side PUBLISHER role tokens; 1-hour expiry
Blockchain	Solana Devnet + Umi + Metaplex Core	Sub-cent transaction cost; soulbound NFT audit model
Styling	CSS + Tailwind	Canvas waveform, glassmorphism UI, mobile-first layout

8. Known Constraints and Engineering Trade-offs

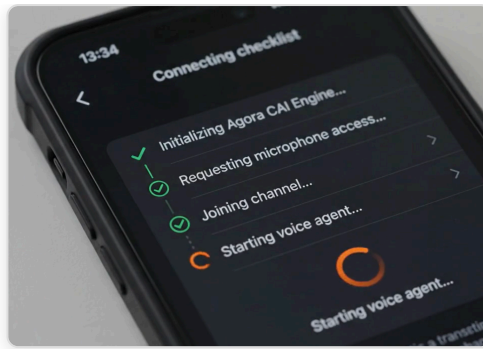
Constraint	Root Cause	Current Mitigation	Future Improvement
0.48s VAD silence threshold	Deepgram/Agora VAD minimum	silence_duration_ms: 480	Implement incremental streaming VAD
End-to-end latency 1.2-2.5s	STT + LLM + TTS in serial	Low-temperature LLM for fast decoding	Streaming LLM output directly to TTS
HFP 16kHz audio quality	Bluetooth specification limit	AEC + ANS + AGC at WebRTC layer	USB-C microphone bypass evaluation
iOS AudioContext suspend	Known Safari regression	visibilitychange resume hook	Persistent Service Worker + WakeLock
No offline fallback	Cloud-only CAI engine	Accepted for MVP scope	Edge-deployed Whisper.cpp for offline STT
Token 1-hour expiry	Agora RTC token TTL	idleTimeout 300s keeps agent alive	Auto-refresh token before expiry window

9. Live Demo Evidence

Note on Deployment Status: The following screenshots are from a fully operational instance of WaveLens Lite running on local hardware. The system is **not yet deployed to a cloud server** — this phase validates the complete hardware-to-cloud pipeline on physical devices before production deployment.

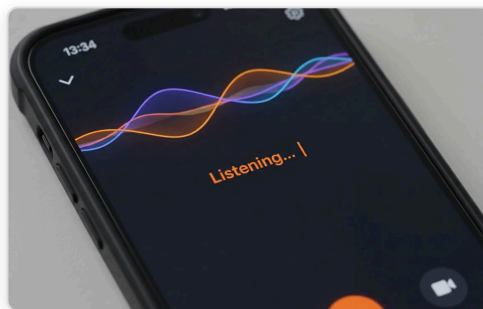
Screenshot 1 — Initial Loading State

The 5-step initialization pipeline: Engine — Microphone — Channel — Agent — RTT



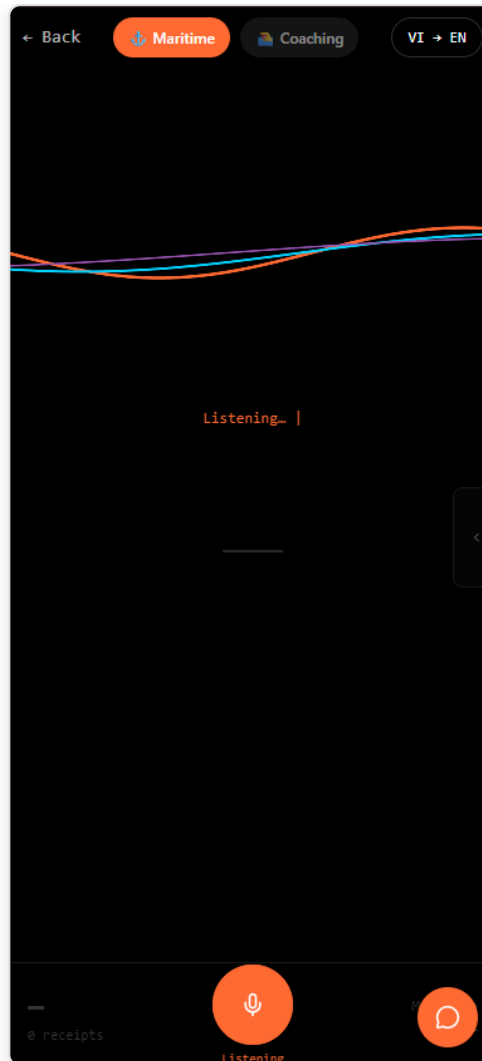
Screenshot 2 — Session Start

Channel joined, Agora agent starting, microphone track published to SDRTN



Screenshot 3 — Live Listening on Mobile

Microphone active, waveform animated in orange, Listening indicator active at bottom



Screenshot 4 — Hardware Field Configuration

Bluetooth earphone worn as intended for real-world use — paired directly to the mobile device, providing clear audio input and synthesized translated speech output during live sessions



10. Conclusion

WaveLens Lite represents a practical, production-validated approach to a real industrial problem: the communication barrier between Vietnamese port workers and international ship crews. The architecture makes principled engineering trade-offs at every

layer.

- **Reliability over features:** The state machine and connection guards prioritize zero-crash operation.
- **Privacy over convenience:** Blockchain audit uses hash-only commitment, not raw transcripts.
- **Resilience over optimization:** VAD thresholds and hardware constraints are tuned for a 90 dB industrial floor, not a quiet office.

The full pipeline — from Bluetooth earphone microphone input, through Agora CAI cloud processing, to synthesized English speech in the earphone — is validated and operational on physical hardware. The remaining steps before port deployment are: cloud hosting on Vercel, an `https://` domain (required for mobile microphone access), and VAD latency fine-tuning for long-form command translation.

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